

UTAP - Unique Topologies for Acoustic Propagation: Designing Algorithmic Waveguides for Sensing in Interactive Malleable Interfaces

Jan Rod^{2,*}, David Collins^{2,3,*}, Daniel Wessolek²,
Thavishi Illandara², Ye Ai², Hyowon Lee², Suranga Nanayakkara²

¹ Keio University, Graduate School of Media Design, Tokyo, Japan. janrod@kmd.keio.ac.jp

² Singapore University of Technology and Design, Singapore
{david_collins, thavishi, daniel, ye_ai, hlee, suranga}@sutd.edu.sg

³ Massachusetts Institute of Technology, Cambridge, MA, USA

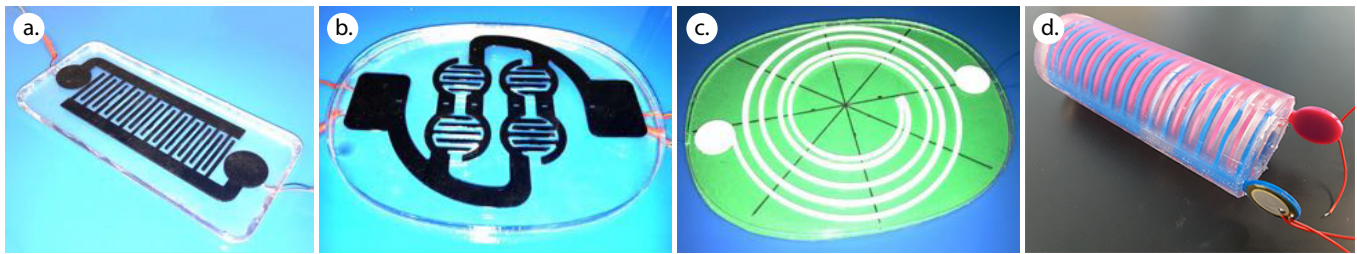


Figure 1. Systematic approach towards building malleable tangible interfaces: Using algorithmically produced acoustic waveguides, our interfaces can sense deformation positions, different force levels, as well other deformations including bending and twisting. In this figure we show prototype interfaces tested and evaluated within this paper.

ABSTRACT

Construction and sensing within malleable interfaces is usually limited by a number of constraints. Building the interface from diverse combinations of conductive and nonconductive soft materials, such as fabrics or foams combined with various sensors, complicates the manufacturing process and offers limited options in shaping. In this paper we propose “*Unique Topologies for Acoustic Propagation*” (UTAP), a novel approach for algorithmic design of malleable tangible interfaces. A fundamental feature of our approach is the implementation of algorithmically generated topologically distinct lattices that, attached to piezoelectric (PZT) transducers, allow us to sense and recognize changes in a modulated acoustic signal on deformation and classify it into different interaction states. Our systematic approach to manufacturing malleable interfaces opens possibilities to design shapes that allow implementation in a wide range of potential use cases. We demonstrate the UTAP approach on multiple interfaces assembled using laser cut and 3D printed lattices in conjunction with silicon compound moulding. Finally, we present a technical evaluation of our method based on studies of four distinct interface designs, assessing performance in sensing and localising simple deformations such as pressing on sin-

gle and multiple spots, as well as different force levels and actions, including bending and twisting.

Author Keywords

Malleable interfaces, acoustic sensing, deformable objects

ACM Classification Keywords

H.5.2. Information Interfaces and Presentation (e.g. HCI): Input devices and strategies

INTRODUCTION

Malleable interfaces offer new interaction possibilities where tangible objects can take advantage of the extensive manipulatory abilities of human hands [4]. Sensing and differentiating deformations in malleable interfaces has, however, been a challenge and often imposes limits on the potential such interfaces offer. In this paper we present a novel, systematic technique for building soft, malleable tangible interfaces containing algorithmically-generated topologically distinct lattices with unique acoustic properties that allow for sensing of unique acoustic responses with different types of deformation (see Figure 2 for system structure overview). These lattice structures are affixed to piezoelectric (PZT) transducers that serve both as emitters and sensors of acoustic frequency sweeps optimized for given combinations of lattice dimensions, type of materials used and frequency responses of the transducers. Since different interactions produce unique changes in the acoustic signal profile, we classify and subsequently recognize them using machine learning.

*These authors contributed equally to this work.

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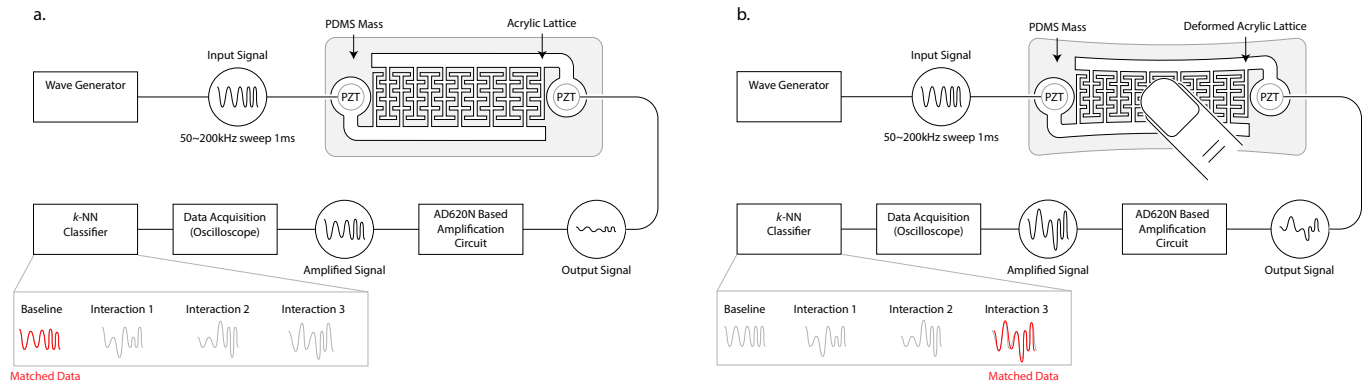


Figure 2. System architecture presented in a use case with one of our interfaces in two states. (a) Baseline state - no interaction, and (b) pressed, deformed state where output signal is matched to one of pre-existing states system was taught during supervised machine learning.

We tested UTAP approach on four different interface types (see Figure 1a-d) moulded from polydimethylsiloxane (PDMS) with laser cut acrylic and 3D printed lattice structures. The results show that our technique is precise in detecting single and multiple simultaneous deformation locations, up to five different applied pressure levels and bending/twisting deformations of the entire interface.

We envision usage scenarios where tangible interaction is desirable and current sensing technologies (e.g. computer vision, pressure and flex sensors, capacitive touch sensing and radar) do not provide optimal solutions in the suitable combinations of sensing capabilities (force, location, duration), tangible properties and freedom in creating desired form that fit a particular use case. This sensing approach is readily scalable, with any number of interaction states being detected with only one single-channel sensing element.

The contribution of this paper consist of:

- Description of a novel approach to designing algorithmic waveguides for acoustic sensing in malleable interfaces - *Unique Topologies for Acoustic Propagation (UTAP)*
- Proof of concept *implementation of various malleable interfaces* moulded from a silicone elastomer with embedded topologically distinct acrylic lattice structures
- Technical *validation of our sensing method* for various deformation using four different interfaces

OVERVIEW OF RELATED RESEARCH

Sensing Techniques on Hard Surface

Sensing interactions between machines and humans is one of the most prominent areas of research in the field of Human Computer Interaction (HCI). A number of approaches have been demonstrated for sensing touch interactions on hard surfaces, including optical sensing [22, 5], capacitive sensing [35, 36], time domain reflectometry [47], acoustic sensing [29, 8] and various types of pressure sensing [37], both on displays as well as other types of hard surfaces that serve as Human Machine Interfaces (HMI). A similar set of techniques is available for tracking of hand motions in the air where no surface is present at all, where the computer recognizes hand gestures, and can be built around use of optical

sensors with both high [1] and low spatial resolution [48], laser rangefinding [10] or, recently, radar waves [2]. A few of these approaches have been applied to sensing techniques used in malleable interfaces, together with various implementations of pressure sensors and flex sensors [40].

Sensing on Malleable Interfaces

Elastic, soft and malleable materials are becoming increasingly popular and are used to build interfaces that are tangible and can support diverse types of rich interactions including squeezing for physical therapy applications [3, 6, 30] and enhancing interactions with plush toys [16]. Other explorations are looking into how to enhance interactions with existing systems and devices [24] as well as envision completely novel ones [15, 45, 49].

Construction and sensing within malleable interfaces, however, is usually limited by fabrication constraints. Building the interface from combinations of conductive and non-conductive materials, such as fabrics or foams, complicates the manufacturing process [32] and offers limited options in shaping the resulting interface. Using large number of pressure sensors placed on the surface [41] or embedded into the a soft structure during moulding [40] affords substantial sensitivity, but requires en-masse integration of sensor components and multichannel processing. Some promising methods limit the spatial extent of sensing, usually into 2.5 dimensions, such as camera based optical sensing [14], though this requires it to be placed on (or part of) a transparent hard flat surface, also a limitations of some other approaches, including those utilizing ferromagnetic fluids [25, 23], or piezoelectric foils [39, 38].

Applications of Acoustics

Acoustic sensing has been implemented in previous HCI work to understand types of interactions [29, 42], augment mobile phones with tangible interfaces [27], or to discover surroundings of a given device using ultrasound of [28]. Distinguishable acoustic properties of simple interactions have been researched for passive understanding of scratching and touching on surfaces [18, 21], as well as recognizing an object or part of a touching finger [19], or even interactions on

human skin [20]. However, the majority of contemporary research utilizes passive acoustics, where the possibilities of active acoustics and its properties for enhanced sensing of different interactions remains relatively unexplored. Some recent work shows that active acoustic sensing approach can substitute pressure sensing [34], or understand different configurations of object assemblies and different user interactions [33]. We acknowledge that our sensing pipeline is analogous to the Touch & Activate system [33], in as much as it also uses frequency sweeps, realtime Fast Fourier Transformation (FFT), analysis of changes in the frequency domain, and machine learning to recognize different states. Our principal contribution lies in formalising the integration and design of the topologically distinct waveguides that allow for custom designed soft interfaces supporting a variety of interactions, specifically in malleable interfaces. Our approach also overcomes some of the limitations of Touch & Activate system, which was limited to sensing on hard surfaces and with non-symmetric objects. By integrating a waveguide material within a malleable interface, we permit the use of soft materials (with higher acoustic attenuation constants) and the decoupling of the interface shape from the ability to sense interaction locations. While our UTAP approach could sense specific interaction locations on a soft sphere surface, for example, Touch & Activate could not.

Active acoustic methods outside of the HCI field have been widely used in a number of different sensing and actuation technologies over a number of length scales, where acoustic fields are capable of imaging and/or manipulating objects from cells [12, 11] to macro-scale structures [31]. These dimensions can be accessed via the wide range of acoustic frequencies that can be generated, with the acoustic wavelength given by

$$\lambda = c/f, \quad (1)$$

where c is the sound speed in the material and f is the applied frequency. For readily generated frequencies on the order of 10^2 – 10^9 Hz and materials with $c \sim 10^3$ ms⁻¹, this corresponds to wavelengths meters to microns. This high degree of resolution, combined with the ability for acoustic fields to interact with structures embedded within the bulk of a material, has made acoustic fields useful for sensing applications including non-destructive testing [26] and ultrasonic imaging [13]. While 2D or 3D imaging using acoustic fields typically requires an array of combined mode acoustic actuators/sensors with sufficient temporal resolution for time-of-flight measurements, we propose localization through the use of spatially varying waveguide structures embedded within the sensing material; when these structures vary non-symmetrically, position can be inferred from the signal received at a single sensing transducer along the axes where symmetry is broken by the waveguide elements.

SENSING METHOD

The sensor localization in interfaces built using UTAP method operates as such: (1) a broadband transducer vibrates (2) an acoustic waveguide that is used to preferentially conduct acoustic energy along transducer-side structure elements, whereafter (3) acoustic energy preferentially transmits

to sensing-side elements at local resonance conditions and (4) the acoustic energy continues to propagate through sensing-side waveguide material and (5) couple into the piezoelectric sensing transducer. A simple operational amplifier circuit can be used to enhance sensing sensitivity. Preferential propagation through the waveguide material occurs when the acoustic attenuation coefficient in the waveguide material, α_w , is higher than that of the intervening medium α_m and there is a significant acoustic reflection coefficient between the two materials (the acoustic energy is largely contained within the waveguide, as is the case with the hard acrylic embedded in a deformable silicone elastomer used in this work). The attenuation and reflection coefficients are given by

$$\alpha = \alpha_0 \omega^\gamma, \quad (2)$$

$$R = \left(\frac{z_2 - z_1}{z_2 + z_1} \right)^2, \quad (3)$$

where the acoustic displacement follows the form $\xi = \xi_0 e^{-\alpha x}$, ω is the angular frequency, α_0 and γ are material specific factors (though $\gamma = 2$ for typical non-viscoelastic materials including the acrylic waveguide material used here), and z_1, z_2 are the acoustic impedances of the pre- and post-interface mediums, with $z_i = \rho_i c_i$. Acoustic transmission between nearby waveguide structures is optimized at resonance (with $f = f^*$), a condition that occurs when the compression wave emanating from the transmitting waveguide element coincides with and reinforces that reflected from the receiving one, specifically at integers of half-wavelengths (i.e., $N = 1/2, 1, 3/2 \dots, f^* = c/(N\lambda)$).

A straightforward implementation embedded resonating elements utilizes structures based on chirped interdigital transducers (IDTs), where the spacing between alternating transmitting and sensing IDT elements varies from one end of the structure to the next. As each of these spacings corresponds to a specific frequency as per Eqn. 1, localization across the spatial extent occurs in the frequency domain, as shown in Figure 3a and 3b. The range of appropriate IDT element spacings for a given frequency range bounded by f_{min} and f_{max} is given by

$$\begin{aligned} \lambda_{max} &= c/f_{min}, \\ \lambda_{min} &= c/f_{max}. \end{aligned} \quad (4)$$

These λ_{max} and λ_{min} values indicate the maximum and minimum realizable IDT spacings, respectively. Though multiple half-integer values of λ are capable of generating resonant conditions between oscillating elements, the smallest one (and the one which yields the highest spatial resolution) is $N = 1/2$: accordingly, the IDT finger spacings should vary between $\lambda_{max}/2$ and $\lambda_{min}/2$. To generate a structure whose spacings vary along a linear axis (i.e. Figs 1a,d) the spacings simply vary from one IDT finger-pair to the next with a defined increment. Starting from the finger position $x_0 = 0$ at the end with the largest IDT spacing and given an increment value ϵ , the centre positions of subsequent fingers are given

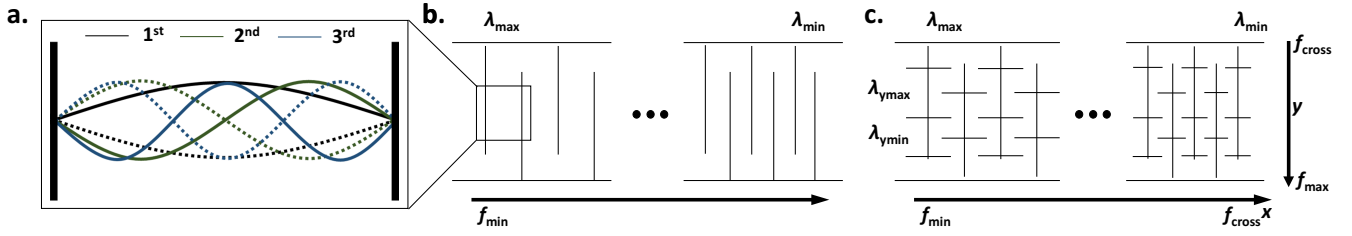


Figure 3. Functional principle of UTAP. (a) The transmission of acoustic energy is maximized when the distance between the transmission and sensing elements corresponds to an integer number of half wavelengths for a given frequency - the first, second and third harmonic resonance modes are shown. (b) Smaller wavelengths, represented by λ , correspond to higher frequencies. By varying λ between transmitting and sensing waveguide elements across a spatial extent, perturbations in the spatial domain can be mapped to the frequency domain, permitting the localization of interaction when a user deforms the waveguide elements. (c) This approach can be extended into two dimensions, where one of two transducers emits a broadband acoustic signal through a stiff waveguide material that is sensed by the other transducer affixed to a interleaved sensing waveguide. Each x, y position corresponds to a unique set of frequencies: the x position to $f_{min} < f_x < f_{cross}$ and the y position to $f_{cross} < f_y < f_{max}$.

by

$$\begin{aligned} x_1 &= x_0 + \lambda_{max}/2 - \epsilon + w/2, \\ x_2 &= x_1 + \lambda_{max}/2 - 2\epsilon + w/2, \\ x_3 &= x_2 + \lambda_{max}/2 - 3\epsilon + w/2, \\ &\dots \end{aligned} \quad (5)$$

until $x_n - x_{n-1} \leq \lambda_{min}/2$, where w is the width of the waveguide finger elements and n is the last finger index. To find the value of ϵ appropriate for a given length L of the sensing device that spans finger-spacings in the range of λ values, a simple script can be composed that increments the ϵ value from an arbitrarily small value (i.e., 1e-6 m) until $x_n \leq L$. When the material containing the waveguide structures is deformed by user interaction at a given location, the position information is encoded in the change in acoustic transmission coefficient at acoustic frequencies that transmit across the deformed elements. Though this is the theoretical optimum sensing range, it is somewhat indicative, however, as localization only requires that each point on an interaction surface has a unique fingerprint of acoustic transmission from the transmission to sensing side in the frequency domain, similar to capacitive sensing work [35].

This can occur outside the indicated IDT spacing range in Eqn. 4, with some acoustic energy transmitting/reflecting due to second-order effects including off-resonance transmission, acoustic interactions within the waveguide material, acoustic reflections with the spatial extents of the bounding media and, of course, deformations induced by user interactions. Nevertheless, Eqn. 4 offers a good first-order guide to the relationship between waveguide element spacings and the appropriate frequency range applied, and where the sensing is more robust and the response in the frequency domain more easily interpreted when the frequency sweeps occur within the range in Eqn. 4.

This method can utilize any surface manifold by integrating topologically distinct structures along its boundaries, such that the fingerprint of acoustic behaviour is unique to each location. However, in the example given in Eqn. 5, the structure spacings vary linearly between the maximum spacing and the minimum one; while only a unique, non symmetric environment is required for position sensing, a linear gradation is more likely to be fault tolerant (indicating a nearby position

rather than a far away one) than one where the spacing increments occur randomly. In the case of a planar interface for sensing along two dimensions this is readily realized by creating IDT structures, potentially in two-dimensions across the x - y plane, as shown in Figure 3c. Both x and y position sensing are independently retained when there is no overlap in the frequency ranges corresponding to the wavelengths in each direction, and where the crossover frequency f_{cross} simply corresponds to the a coordinate shift from the x -dimension to y -dimension in Eqn. 5.

IMPLEMENTATION

A schematic overview of our system is shown in Figure 2. A PZT transducer is affixed to the IDT transmission and sensing lattice components. The lattices are completely encompassed by a malleable material. The actively powered transmission transducer is actuated by a frequency-swept signal that is generated by a function generator. The acoustic signal that is captured by the sensing-side transducer is captured with an oscilloscope and uploaded in real-time to a computer for analysis and classification, using a supervised machine learning algorithm. Due to the differences in the acoustic profile that occur when the lattice is deformed, the system can recognize various (prior trained) states. Subsequently, these interaction states can be mapped to particular functions on a target system.

We have built four interfaces using PDMS moulding with lattice structures to test our sensing method. The lattice structures were produced by two different techniques. The first technique uses a laser cutter, is simple and can be replicated using a hobby equipment, while the second one, using additive manufacturing, is geared towards building more complex structures. Using the first method, we also demonstrate various approaches towards designing the lattice structures that confer spatial localization, namely a 1D grid, specific malleable regions, as well as continuously converging spiral IDT elements.

For the purpose of evaluating feasibility of algorithmic production of the IDT lattices, we created a script written in Matlab programming language that generates spacing coordinates to define a unique position in three-dimensional space for each of the topological features following Eq. 5 within given parameters: generating the point cloud in one, two or three

dimensions, overall size of the interface and initial spacing parameters related to invariable characteristics of used material (fixed thickness of the acrylic). These coordinates are written into a CSV file. Subsequently, we import these coordinates into a CAD software (Autodesk Inventor) and use them to drive placement of the pre-designed IDT fingers (cf. Fig. 4). In this early stage, we finalize the outer shape of the interface manually to give it a desired shape. To demonstrate the versatility of the approach, we have constructed four diverse interfaces, where Fig. 1a shows the simplest linear deformation sensing. As more closely spaced elements yield greater position selectivity, the range of spacings chosen (centre positions 3.75 mm to 5 mm, spacing between elements 1.75 mm to 3 mm with a $w = 2$ mm finger width) were the smallest routinely realizable given the fabrication procedure (where elements may shift slightly during elastomer curing). These spacings, given fundamental mode resonance ($N=1$, half-wavelength spacing) gives values of $f_{min} = 72$ kHz and $f_{max} = 123$ kHz as per Eqn. 4, well within the actuation and sensing range of working frequencies of our piezo elements. To design a system with a representative length L within 100 mm given these values for $\lambda_{max}/2$ and $\lambda_{min}/2$, Eqn. 5 can be used to define the waveguide finger spacings along the x -axis, and where a simple script was utilized to increase the increment (ϵ in Eqn. 5) between finger pairs until $x_n \leq L$, where x_n is position of the last element. In the sensing device here this process gives an increment of 0.06 mm. The interface in Fig. 1b demonstrates the versatility of variable waveguide spacing where the lattices can be shaped to form deformable areas mimicking functionality of buttons. Interface Fig. 1c then further expands on the versatility of the embedded waveguide approach using converging spirals, where variable spacing is accomplished via continuously converging elements rather than discretely incremented positions. The interface Fig. 1d is then an attempt to evaluate three-dimensional shape that is better geared towards supporting bending and twisting interactions, and is indicative of the range of interfaces that can be produced using this approach. These design choices were made to show versatility and robustness of our approach in this early stage of our research.

For the transducer instrumentation we experimented with PZTs with 27 mm diameter from different manufacturers to test the universal applicability of our approach, all inexpensive designs (< 0.20 US\$) used in, for example, alarm buzzers. Different piezo transducers have varying amplitude responses for given applied frequencies, which needs to be reflected when defining appropriate frequency sweeps for each specific application and material combination. We have tested the frequency responses for each batch of PZTs that we acquired and selectively worked with those that had good response in a higher frequency spectrum. In our technical evaluation, we use frequency sweeps in the 50 kHz to 200 kHz range (with a 1 ms sweep period), where we achieved best frequency response with our chosen PZT transducer and which covers our frequencies dictated by the algorithm used for producing the IDT interfaces. This range is well above the range of human hearing (approx. 50 Hz to 20 kHz). Though specialized transducers should be able to read-

ily generate higher frequency displacements more effectively, the use of non-optimal transducers in the case here demonstrates the robustness of this method and its applicability to low-cost position sensing.

Interface Construction

We present and evaluate four lattice designs (three produced using 2D laser cutting and one produced using additive manufacturing), some of which enable malleability of the whole surface others with specifically designed malleable regions (Cf. type (a) and type (b) in Figure 1).

Our tested IDT lattice structures are fabricated from acrylic using simple 2D laser cutting as well as additive manufacturing for more complex 3D interfaces. For 2D laser cutting, we used 2 mm material as an attempt to compromise between stiffness and lateral acoustic wave propagation; thicker material results in a higher acoustic amplitude at the sensing transducer, while a thinner material is more readily deformable (and can be used for smaller scale interfaces - i.e. watch size).

Prototype IDTs fabricated using additive manufacturing (VeroCyan RGD843 and VeroMagenta RGD851 materials on Stratasys Objet 3D printer) have the same 2 mm thickness to achieve comparable acoustic conductivity at the desired scale of the interface. We subsequently attach two piezo transducers directly to the transducer pads on the IDT using epoxy glue, resulting in good acoustic coupling between the transducer's brass disc and the acrylic structure. Using simple moulds, we encase the whole structure in PDMS curable elastomer material (sylguard 184, Dow Corning, USA). First, we pour and cure a thin 1 mm layer of PDMS that will form the top of the interface. Once the initial layer is cured, we insert and position the lattice and pour another layer of PDMS over it (See Figure 4 for a construction diagram). The resulting interface can conceivably be much larger (or smaller), but here range in size between that of a smartphone and a tablet computer. For the 3D printed lattice, we have designed a specific mould that held both IDTs in correct position to assure accurate spacing along all three axes. Based on the lattice design implemented, finalized interface can be pressed, bent and twisted depending on desired interaction.

Acoustic Sweep Sensing and Interaction Recognition

Due to the specific properties of each construction methods and also specific response frequencies of the PZT elements, we explore the responsive frequencies of each interface after the construction and discover frequencies that give consistent specific responses to different interactions. Subsequently we use this data to construct the acoustic frequency sweeps. For example, the interface built using laser cut lattice and PDMS moulding, given the acoustic properties of acrylic (velocity ca. 2600 m/s and acoustic impedance of $3.1e6$ Rayls) and PDMS (velocity ca. 1076 m/s and acoustic impedance of $1.1e6$ Rayls), lattice features on the order of cm to mm and our transducer element response, we utilize frequencies within a range of 50 to 200 kHz.

The output signal is then amplified using a custom circuit built around two AD620 instrumentation amplifiers. Subsequently, we capture a fixed window of data points using an

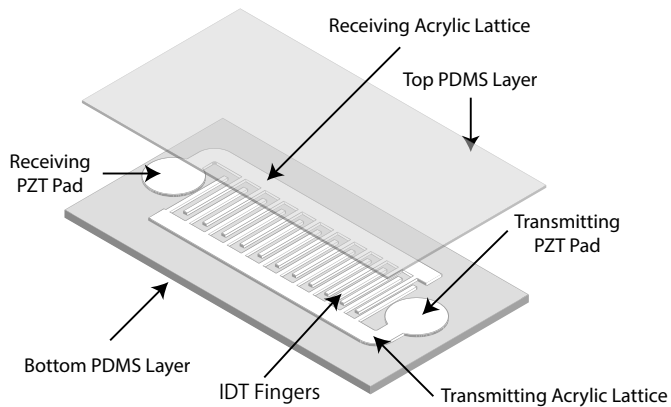


Figure 4. Overview of a malleable interface construction using laser cut IDT lattice and PDMS moulding.

oscilloscope, from where it is transferred to a computer. We then perform an FFT analysis on selected frequency range resulting in 750 data points in each sample we further use for interaction recognition. Interaction recognition is achieved using simple k -Nearest Neighbour (k -NN) supervised machine learning algorithm from WEKA machine learning toolkit [17] with $k = 1$ to train the system acoustic profile of a baseline state with no interaction happening and various states where certain interaction occurs. Both in test and live setup, we observe deformations in the acoustic spectrum and match them with previously trained states to recognize different types of deformations.

Our system can also to be used in live environment, using continuous capture of the acoustic signal and analysis on the fly using the same (k -NN) algorithm. The current implementation of this system uses a fixed data window capture from an oscilloscope tethered to a computer and has a recognition frequency around 10 Hz due to the equipment limitations (mainly upload delay from our oscilloscope and delay introduced by performing the FFT analysis on the fly), however due to the material resistance and relaxation timescales in a thick material, individual interactions occur over timescales longer than 1/10s, thus this speed is sufficient for our early implementation and evaluation of the UTAP approach. Subsequent implementations may make use of specialized FFT microchips to perform higher refresh-rate sensing.

EVALUATION AND RESULTS

To evaluate our sensing technique, we have conducted separate experiments using three types of interfaces constructed to measure the accuracy in sensing different types of deformations. Specifically, we have measured ability of the UTAP approach to sense individual *local deformations* (i.e. pressing with human finger and other objects), *multiple local deformations* occurring at the same time (i.e. pressing with multiple fingers), five different *force levels*, as well as *bending* and *twisting* actions.

Evaluation Methodology

For each of the experiments, we create our training set capturing 100 samples of data from the oscilloscope for each of the

pre-defined states. A state is defined as a deformation, such as pressure applied by a human digit on a certain area of the interface, or a bend performed by a user holding the interface in a specific way. This was also performed for the baseline state when the interface is not deformed at all. We recreate each of the states and capture a testing data set with 50 samples of each state. All of the interactions are performed by one of the researchers actuating the designated interaction area to first collect the training samples and later presses on the same areas to collect the test samples. Provided the deformations are happening in the same location, the precision will not change from user to user, or even between a human digit and other similarly shaped object deforming the interface with similar force, since the deformation is sensed by varying frequency response of the waveguides. Particular type of actuator used is irrelevant as far as it produces given deformation. This was tested by authors, but is not part of results reported in this paper. To prevent loss of accuracy through undesired acoustic interactions between the interfaces and the surface they are placed on, we have placed each of the interfaces onto a styrofoam block to acoustically isolate them where necessary. Data sets are captured to a computer using oscilloscope and FFT function is performed on each data sample. The resulting FFT features (750 data points) are then compared using the k -NN algorithm with $k = 1$. We present results from four separate studies carried out on four different interfaces seen in Figures 1(a), (b), (c) and (d). Respective state configurations are described within the sections. We have focused on evaluating different types of interactions as well as different types of lattice structures.

Results

1-Dimensional Interface

This interface shown in Figure 1(a) is built to detect deformations using a 1-dimensional lattice structure - i.e. the change in spacing of the IDTs occurs on one axis, allowing us to sense deformations along that dimension. We trained it to recognize 7 deformation regions along the lateral axis of the interface (labeled P1 - P7), as well as different types of deformations: bending upwards (B1), bending downwards (B2), twisting left (T1) and twisting right (T2). Bending states were achieved by supporting the interface with a hard object on each end (B1) and then bending it using a human digit pressing on the middle part, as well as supporting it in the middle and pressing on the outer ends (B2). Twisting was achieved by supporting the interface in diagonal corners and pressing on the unsupported corners to create a lateral left (T1) and right (T2) twist. The results can be seen in the form of confusion matrix in Figure 6(a). The overall accuracy of 92 % points to a high degree of sensed state specificity. Some states were mis-classified as neighbouring states due to relatively higher density of trained regions. Note that the bending and twisting deformations have 100 % recognition rate. The overall accuracy increases with a smaller number of trained interaction states (99 % for three trained locations).

Multipoint Interface

The Multipoint interface (Figure 1(b)) demonstrates the potential to sense deformations in multiple isolated regions. In addition to sensing deformations in each of the regions, our

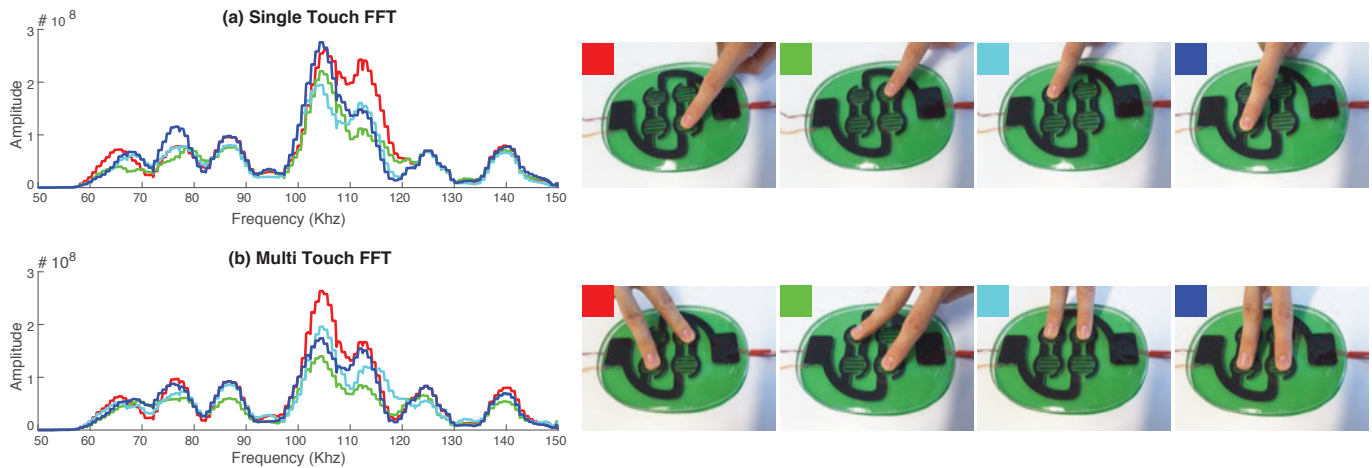


Figure 5. Sample comparison of FFT differences between interaction states. Figure (a) shows differences for single points pressed states as seen in corresponding photographs, figure (b) displays multiple points pressed at the same time. All results are averages of 50 data samples for each state compared to normalized common baseline. Plot cropped to range of 50 kHz to 150 kHz. Note that the PZTs have similar frequency response profiles, thus the profile of each interaction state may also appear similar.

aim was to use the distinct regions for straightforward evaluation of different combinations of deformations within the defined regions (labeled P1 to P4), as well as different pressures (F1 to F2), where each multiple deformation (i.e. P1P3 means simultaneous deformation of *point 1* and *point 3*) or pressure level is defined as a dedicated state (i.e. P1F1 means *point 1*, *force level 1*). Results can be seen in Figure 6(b) with overall accuracy of 95.7%. In addition, we present averaged normalized FFT curves for some states in Fig. 5 for purpose of illustrating the differences in frequency spectrum. Interestingly, our system can recognize five different force levels (F1 to F5) in one region of the interface with an accuracy of 90% (cf. Figure 6(e)). For this particular study, force levels were consistently produced using a digit analogue (a rounded object with diameter of 10 mm resembling a human digit) with applied mass/force increments of 0.8 kg / 78N.

Spiral Interface

The spiral interface seen in Figure 1(c), is an attempt to evaluate a lattice in the form of 2 converging spirals. Similar to the other lattice structures, this interface can yield deformation localization through a distinct fingerprint of local topological features, here as the effective distance between transducer and sensing elements continuously changes from the outside to the inner part of the sensing interface rather than the discrete changes in the other tested interfaces. We have trained it with 11 different deformation points (labeled P1 to P11) distributed along the spiral converging inwards to the center of the interface. The results can be seen in the form of confusion matrix in Figure 6(c). Here we can observe one state which tended to be misclassified, arguably due to lack of significantly pronounced geometrical features. Overall results, however, were nonetheless indicate 94.8% accuracy.

Interface Using Lattice Fabricated by Additive Manufacturing

The spiral interface seen in Figure 1(d), is an attempt of basic evaluation of a lattice fabricated using additive 3D manufacturing. Oval shape of the interface is geared towards evaluating bending and twisting. We have defined two bend states,

up and down (B1 and B2) as well as twist clockwise and counterclockwise (T1 and T2). The results can be seen in Figure 6(d). The overall result is 90.0% of successful recognition. We can observe that the twisting states, while generally recognized, have significantly lower successful recognition rate. The reason for the lower performance is lesser deformation of the lattice structure occurring during twisting, since the “ribs” of the lattice structure do not displace significantly. The potential strategies to increase twist sensing accuracy would be to design additional features that would increase displacement of the whole lattice during twisting interaction, thus changing the acoustic profile of such interaction in a more significant manner.

Discussion

Our results show that the UTAP approach is precise in detecting different deformation locations and states. The theoretical spatial precision of UTAP is very high with ability to detect deformations at the same time. However, the geometrical properties of lattices used together with a rudimentary method of training the machine learning algorithm decrease the precision in our particular implementation of the system. While the difference in waveguide topology between spiral and rectilinear waveguide elements complicates a direct comparison between multipoint interfaces, a spiral interface is beneficial from the standpoint of being simple to design though may suffer from poor spatial resolution and weak signal due to the continuous (rather than discrete) nature of spacing increment, which limits the robust recognition of discrete states, and the long waveguide transmission path over which acoustic energy is transmitted, especially towards the spiral centre. Interestingly, UTAP can also detect different force levels, which we classify as different states of deformation in the same region. The consistent performance points towards good system trainability and present a solid basis for further exploration of this technique. Some of the practical limitations of UTAP are further discussed in-depth within the following section.

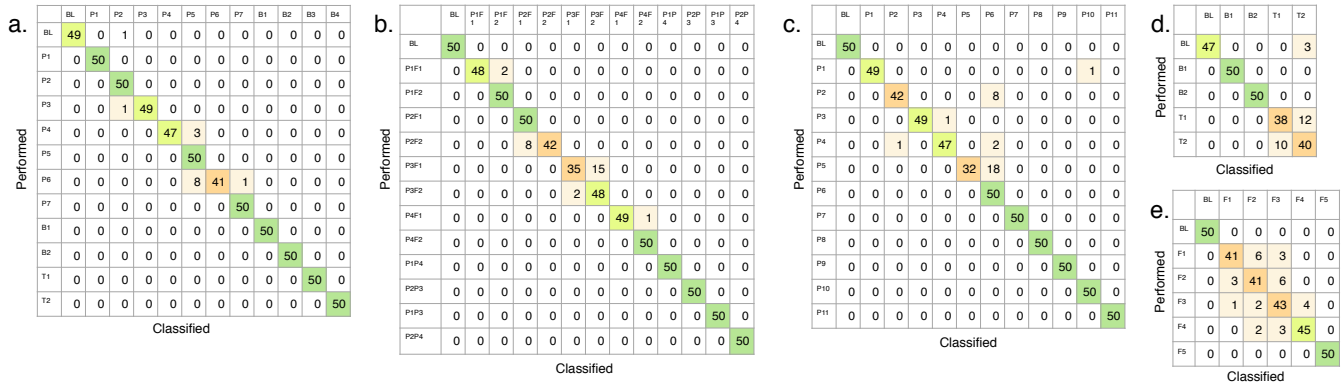


Figure 6. Experiment results: Tables (a) to (e) show resulting confusion matrices of each tested deformation for corresponding interfaces in displayed in Figure 1. Label explanations in the main text, BL = Baseline state.

LIMITATIONS

While in principle UTAP allows for very precise deformations measurement in wide scope of interface shapes and designs, there are some practical and theoretical limitations that impact the method and the interface fabrication. Furthermore, due to the PZT transducers being enclosed in the bulk matter of the interface, UTAP allows for creating interfaces with low instrumentation footprint, even though the interface still needs to be wired to the rest of the instruments, rendering some handheld operations slightly more cumbersome.

General Limitations of UTAP

Though there is no inherent theoretical limit in the spatial resolution that can be realized using UTAP, in this implementation the limited frequency range permitted with the transducers implemented and the laser-cutting manufacturing method used results in detection resolution on the order of the cm-scale features utilized. As the manufacture of smaller components and higher frequency transducing and data recording elements is substantially more cost intensive, this technique is likely optimal for low cost sensing applications that require relatively coarse resolution (with respect to a touchscreen, for example). The sensed state is also sensitive to the conditions during which the training data capture was undertaken; in the case of our setups described herein, the interfaces were acoustically isolated on styrofoam blocks for consistent testing conditions between measurements. Measurement of deformation states does not necessarily require this, though future work on the recognition engine is needed to automatically update and recognize different states irrespective of boundary conditions. The spatial limits of the sensing area are a function of the spatial sensitivity required; lower frequencies result in correspondingly lower attenuation and greater signal transmission, but require larger waveguide spacings to robustly sense discrete states. Hence, larger interfaces have an inherent tradeoff with spatial resolution, though the particular limits for a given implementation are a function of practical considerations including sensitivity of the sensing transducer and the applied power. Using our systems here as a first-order example, however, we have shown that cm-scale spatial resolution is possible in 10 to 20 cm scale interfaces.

Materials and Manufacturing

Acoustic attenuation occurring in soft materials negatively influences the signal strength and the possible distances between the two IDT parts of each lattice structure (i.e. if the lattices are too far, the signal will be excessively attenuated). Depending on the strength of the signal that we inject into the interface, our method is suitable for hand-held scale interfaces. Increasing the size demands for stronger signal injected into the interface as well as stronger amplification of sensed signal. However, unlike other similar approaches using acoustic sensing, employing UTAP to design and fabricate custom lattices allows for more control over shape and distribution of the materials with different acoustic properties (and different elastic moduli) - e.g. embed the IDTs which have better acoustic conductivity and lower attenuation coefficient into a “core” of the interface and have a larger mass of soft malleable material on the top, which would still allow to sense deformations but on a significantly larger scale - such as pillow or plush toy.

The cheap, affordable PZT transducers that we use in this proof of concept have variable frequency response across the acoustic spectrum that often differs from one PZT transducer to another. Prior to specifying the actual frequencies for the sweep, we test the frequency response of the transducer pair to see in which frequencies they perform well.

Real Time Interaction Sensing

In our current configuration of the real time sensing the speed of the recognition is around 10 Hz. This limitation is given by limitations of data capture and transfer speed from oscilloscope to the computer as well as FFTs performed on each data window which is later cropped to 750 data points that are used as input for the WEKA machine learning. The real time sensing can be improved to perform at higher frequency (around 50 Hz) using real-time hardware based FFT digital signal processing and implementing faster machine recognition pipeline, such as Sci-Kit Learn library for Python.

FUTURE WORK

Our system at present serves as a proof of concept for our UTAP method and demonstrates some early approaches to

building malleable interfaces. There is a number of improvements we plan to address in continuation of the work, improving the system setup to increase speed recognition, as well as exploring the potential for an automatic, algorithm driven generative method for creating the lattice automatically for any given 3D shape. While foundations for this work are laid out in this paper, additional work is needed to optimize features and properties of the IDT structures and to explore the possibilities this research opens.

Our current system implementation works in real time, however with a slower recognition rate that is sub-optimal for real use. In the future, we expect to improve system hardware pipeline, integrating it into standalone, wireless, inexpensive battery powered module that can be configured for a wide variety of interaction interfaces, which would allow us reduce the form factor of the system to allow for fabricating standalone interfaces. Additional configurability of custom built hardware also allows to experiment with switching cycles of the PZT elements between transmitting and receiving. In such configuration, it is possible to acquire deformed acoustic signal from both IDT structures in one cycle (i.e. in first half-cycle, PZT A transmits a signal and PZT B receives, in the second half-cycle PZT B transmits and PZT A receives, creating symmetric deformation profile) and thus with more information, increase the precision of the system.

We plan to explore a number of potential applications, building specialised interfaces across different scales for use cases we have mentioned earlier, such as custom made rehabilitation devices and testing them for real rehabilitation scenarios in post-traumatic hand/body rehabilitation or post-stroke rehabilitation. We also plan to take popular interaction strategies and modalities used today in our day-to-day use of IT devices, and explore how these could be augmented, enhanced or even replaced when the malleable sensing material we propose in this paper is extensively adopted in their design.

Furthermore, there are other interesting trajectories to explore, such as implementation of actuation properties into the interface, or implementing light-conductive elements that would allow the interfaces to be infused with controlled light to complement the sensing functionality to turn our interfaces to outputs, as well as inputs. Our IDT structures would then gain another functionality as light conductive elements, similar to 3D printed conductive light tubes [9, 46]. Multi-material additive manufacturing methods are suitable for such enhancements. Looking further ahead, we can imagine our method to be used to and sense interactions in greater custom shaped surfaces, such as 3D artificial skin.

POSSIBLE APPLICATIONS

Our technique has multiple potential applications. Soft tangible interfaces provide additional modalities to hard touch-enabled surfaces and enable various types of interactions that take advantage of highly advanced motoric properties of human hands.

Since our method of interface construction is universal, it can be used for specialized single purpose applications to support specific, very narrow and specialised interactions, as well as

to build more generic ones, that can be used and re-trained repeatedly for different purposes. For example, we envision that our method could be used to construct interfaces that require tracking of specific interactions in rehabilitation, such as hand injury requiring repeated exercises on particular muscle groups and mobility of different fingers (e.g. [44]). Another potential application might be post-stroke rehabilitation that often includes training focusing on regaining motor abilities (e.g [3, 7]) to replace current generation of unsuitable “hard” tangible devices. In a general computing applications, we can devise a supportive interfaces for tasks such as 3D modelling where a soft interface can simulate clay and mirror sensed interactions on a modeled object on the screen (e.g. [43, 32]).

A separate group of possible applications is to enhance user experience of interactive devices with malleable, soft components. The hard, sharp and delicate buttons and controllers on our modern IT devices have much room for improvement in offering interactivity. For example, we envision game controllers with flexible handles, or add-ons to other mobile, hand-held devices such as smartphones and tablets where the casing, grips, handles or even the main body of the devices support additional input mechanisms through more natural and intuitive gripping or squeezing action. Importantly, this scalable approach to interaction with malleable interfaces requires only a single sensing channel to measure and detect several different action states using inexpensive sensing and receiving transducers. In most of these application areas, our approach could offer a significant benefits for cost-effect manufacturing of interactive device.

CONCLUSION

In this paper we have presented UTAP, unique topologies for acoustic propagation, a systematic approach towards designing algorithmic waveguides for building malleable tangible interfaces. As this method achieves localization through algorithmic design of integrated waveguides and measurements in the frequency domain, only one sensing element is required. The use of a cheap, off-the-shelf sensor here demonstrates that this localization technique can be applied more cost-effectively than conventional multi-channel sensing devices. Using algorithmically created acoustic waveguides, our system can sense various deformations, such as pressing, bending and twisting, as well as different force levels, a combination of sensing modalities that are not afforded by comparable techniques. We have demonstrated and tested four different types of the interfaces build using laser-cut acrylic lattices moulded into malleable PDMS structures. Our results show that UTAP is a reliable technique with high accuracy.

UTAP, as presented in this paper, is an initial exploration of this technique and we expect it to significantly expand to investigate the implications for building interfaces of novel shapes, allowing for rich interactions, ultimately taking advantage of endless potential and sensomotoric abilities of our human bodies.

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